

# Global Uncertainty and Exchange Rate Conditions: Assessing the Impact of Uncertainty Shocks in Emerging Markets and Advanced Economies

Helena Glebocki Keefe\* & Sujata Saha†

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## Abstract

Economic and financial conditions respond significantly to increased measures of uncertainty, particularly in economic policy, monetary policy, and financial markets. This research employs the global vector autoregression (GVAR) model to assess the response of exchange rate volatility, exchange market pressure, and real effective exchange rates in emerging markets and advanced economies to shocks in global economic policy uncertainty, US monetary policy uncertainty, US equity market volatility, as well as measures of uncertainty based on the financial stress index and geopolitical risk index. Across all measures, spikes in uncertainty correspond to immediate increases in exchange rate volatility, buildup of exchange market pressure, and a depreciation in the real effective exchange rate for emerging markets. The results are compared to advanced economies, China, and Russia separately, identifying key differences across markets.

Keywords: Global Uncertainty; Exchange Rates; Emerging Markets; International Finance

JEL: F31; F42; F65

Authors have no competing interests.

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\*Corresponding Author. Department of Economics, Fairfield University. Fairfield, CT. [hkeefe@fairfield.edu](mailto:hkeefe@fairfield.edu)

†Department of Economics, Wabash College. Crawfordsville, IN. [sahas@wabash.edu](mailto:sahas@wabash.edu)

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## 1 Introduction

Concerns about the economic impact of heightened uncertainty has been pervasive in the academic literature for years. However, with the onset of the global health pandemic in 2020, the Russian invasion of Ukraine in 2022, and the spillover effects of contractionary monetary policy by advanced economies, the global economy has experienced significant and severe waves of uncertainty unlike those seen in recent history. Furthermore, rapid and ongoing innovations in technologies that are transforming the global economic landscape add to the rising concerns about an uncertain future (Al-Thaqeb and Algharabali, 2019). Unsurprisingly, spikes in global economic uncertainty, policy uncertainty, and financial market volatility all negatively affect economic activity. With greater uncertainty, firms are reluctant to make costly decisions in unpredictable conditions (Bloom, 2014). Consumption spending declines, unemployment rises and credit tends to dry up, all of which exacerbate the negative effects of uncertainty spikes.

From a global perspective, uncertainty shocks may be transmitted through the impact on international financial flows and financial markets, trade, and exchange rates. The research presented in

this paper analyzes the impact of uncertainty shocks on exchange rate conditions in emerging and advanced economies. Economic uncertainty can affect foreign exchange markets in a various ways. Policy uncertainty that impacts foreign capital flows can inhibit the willingness of international investors to invest in both domestic and international markets. When economic policy uncertainty corresponds to uncertainty regarding trade and trade agreements, trade flows will be impacted. Since exchange rates are impacted by both changes in investors' relative demand for foreign currency to domestic currency assets, as well as the demand for foreign goods relative to domestic goods, uncertainty that impacts trade and capital flows will affect exchange rates. Liming (2020) and Chang, Derindag, Hacievliyagil and Canakci (2022) demonstrate that with heightened uncertainty, exchange rates are more volatile .

The research presented in this paper analyzes the impact of uncertainty shocks on exchange rate conditions in both emerging and advanced economies and addresses a critical gap in the literature on the effects of uncertainty on exchange rates. First, by utilizing monthly data from 1996 to 2022 for twenty-eight countries plus the euro-zone, the impact of uncertainty shocks is analyzed in a truly global context. Much of the previous research in this area has focused on advanced economies, such as the G7, or individual emerging markets. This research considers both emerging and advanced economies over a period of time that includes several significant, historical and recent, episodes of uncertainty, such as emerging market currency crises, recessions, Global Financial Crisis, European Debt Crisis, Covid-19 and the Russian invasion of Ukraine.

Second, utilizing the global vector autoregression (GVAR) model is a novel approach to analyzing the impact of uncertainty on the global economy. The GVAR model creates a global system where each country's response to the shock controls for domestic, foreign and global factors. The response of exchange rate conditions to uncertainty shocks incorporates the domestic and foreign country inflation rates, interest rates, and industrial production, as well as global oil prices and uncertainty measures. Therefore, by creating a trade-weighted global system of countries and variables, it provides a more holistic analysis of the impact of uncertainty shocks on exchange rate conditions.

Finally, five different measures of uncertainty are introduced as shocks in the global system. The model utilizes the Baker, Bloom and Davis (2016) measures of global economic policy uncertainty and US monetary policy uncertainty, as well as US equity market volatility. The US-specific measures of uncertainty are included since the US dollar plays a critical role in the global financial system as the most heavily traded currency on foreign exchange markets, the dominant currency held as foreign exchange reserves by central banks, and the currency used to price commodities and settle many cross-border transactions. Therefore, US-related policy uncertainty may have a significant effect on the global financial system, currency markets, and exchange rate conditions around the world. The financial stress index compiled by Puttman (2018), which captures a detailed coverage of financial history for more than a century, and the geopolitical risk index compiled by Caldara and Iacoviello (2022) are also used as measures of uncertainty.

The results indicate that spikes in global uncertainty correspond to immediate increases in exchange rate volatility for both emerging and advanced economies. However, the response of exchange market

pressure and real effective exchange rates differs for emerging and advanced economies. In emerging markets, shocks in uncertainty correspond to a buildup of depreciation pressure in both the exchange market pressure index and in the real effective exchange rate. Although still significant, the response is muted in China, which is not surprising based on the interventions and management of the Chinese Yuan. On the other hand, the response in Russia is larger in magnitude and significance compared to other emerging markets. In contrast, there is a buildup of appreciation pressure in the core currencies of advanced economies for both the exchange market pressure index and the real effective exchange rate. This may correspond to the impact of shocks on international financial flows, as investors flock to lower-risk assets in advanced economies during periods of greater uncertainty, leading to the appreciation/depreciation response observed in advanced/emerging economies, respectively.

Although the topic of uncertainty and its economic impact has been studied extensively in the academic literature, the focus has been on macroeconomic variables, such as unemployment, production, and inflation, or financial market conditions, such as stock market returns and stock price volatility. There are few papers that consider the impact on exchange rates explicitly. Understanding the impact on exchange market conditions is critical, particularly for emerging market policymakers. Excessive volatility or drastic movements in exchange rate values may introduce significant instability in domestic economic conditions for this group of economies, which are more susceptible to the international transmission of macroeconomic shocks due to their institutional quality and level of financial development. Lower levels of development and openness correspond to a more significant response to global shocks (Glebocki Keefe, 2021). Therefore, the findings presented in this paper have important implications for policymakers in emerging market economies and for practitioners in international finance.

The remainder of the paper is structured as follows: Section 2 discusses the theoretical and empirical literature on uncertainty and its economic effects. Section 3 presents the data, model, and methodology. Section 4 presents the empirical results, and Section 5 concludes.

## 2 Literature Review

Uncertainty, in general, has a negative impact on economic activity, primarily due to the inability or unwillingness of firms and consumers to make investment and spending decisions for the future during unpredictable times (Bloom, 2009). Firms await more information before making irreversible and costly investment plans (Bernanke, 1983), while households postpone investments due to lower personal income and respond less to market opportunities, price signals and other incentives (Foote, Hurst and Leahy, 2000). Abel (1983) defined uncertainty as unexpected changes in the economy, and studies how these changes in government policies affect investments by corporations. Furthermore, uncertainty regarding economic policies negatively impacts unemployment, inflation, money demand and financial market conditions. Uncertainty significantly increases risk premia in the financial markets, which in turn increases borrowing costs, decreases productivity, and reduces employment Gilchrist, Sim and Zakrajsek (2014).

According to Caggiano, Castelnuovo and Figueres (2017), during recessionary periods, greater economic policy uncertainty contributes to unemployment volatility, and the effects on unemployment are more significant than previous estimations. The effect of uncertainty on employment is affected through channels of economic policies and international trade. Pierce and Schott (2016) examine the effect of changes in international trade policy on domestic employment and found that industries more exposed to the policy change experience a sharper decline in employment when trade policies change.

Furthermore, Leduc and Liu (2016) found that when uncertainty is high, inflation rates are low, and unemployment rates are high. This in turn leads to a reduced consumption demand and spending, generating demand shocks in the economy. Jones and Olsen (2013) estimate the time-varying correlations between uncertainty, inflation and output, finding consistent negative correlations between uncertainty and output, indicating greater uncertainty corresponds to lower output. However, the correlation between uncertainty and inflation varies, and is positive in the post-1990's, corresponding to increasing inflation during periods of greater uncertainty.

In addition to the effect on employment, inflation and investment decisions, economic uncertainty also affects economic growth. Bernanke (1983), Dixit and Pindyck (1994) and Bloom (2014) found that economic uncertainty affects economic growth negatively. More recently, Balcilar, Gupta and Segnon (2016) found a significant impact of economic policy uncertainty on GDP growth and that data on uncertainty can be used to predict US recessions. Barrero, Bloom and Wright (2017) also argued that an increase in policy uncertainty affects economic growth through the channels of reduced capital and R&D investments. Aastveit, Natvik and Sola (2017) find that monetary policy is less effective in periods of greater uncertainty. Specifically, the impact of monetary policy on investment spending is halved in periods of significant uncertainty. To add to this, Balcilar, Demirer, Gupta and Eyden (2017) argued that economic policy uncertainty in the U.S. lowers the effect of monetary policy shocks in the Euro area using a structural interacted vector autoregressive (IVAR) model. Uncertainty in fiscal policies affects economic activity. Fernandez-Villaverde, Guerron-Quintana, Kuester and Rubio-Ramirez (2015) considered tax and spending processes and established how uncertainties in fiscal volatility shocks affect economic activity adversely.

Uncertainty shocks also significantly affect international trade. Novy and Taylor (2020) demonstrate that international trade is disproportionately affected by uncertainty shocks compared to domestic activity. Firms cut orders of foreign inputs due to higher fixed costs during periods of uncertainty, leading to a magnified response. Gervais (2018) develops a theoretical model explaining the impact on uncertainty on international trade, and finds that an increase in uncertainty leads to an increase in the number of suppliers. Firms adjust their supply chains to purchase a larger fraction of inputs from low trade barrier and low uncertainty suppliers, spreading their input demand more evenly across suppliers. Martin, Mejean and Parenti (2020) find that periods of high macroeconomic uncertainty correspond to less trade, which is driven by the decrease in firm-to-firm relationships in these periods. In contrast, Baley, Veldkamp and Waugh (2020) develop of general equilibrium model with information frictions that demonstrates uncertainty as increasing both the mean and variance in returns to exporting, indicating

that trade may increase or decrease with uncertainty contingent on preferences. They determine that if higher uncertainty improves terms of trade, it will correspond to an increase in trade. Additionally, the role of economic policy uncertainty on international trade is explored by Hassan, Shabi and Choudhry (2018). They found that an increase in economic uncertainty affects international trade adversely and that the impact is asymmetric in a way that the effect of a rise in economic uncertainty is more significant than the effect of a decrease.

Uncertainty will also impact international financial flows, financial market conditions and asset prices. Julio and Yook (2016) estimate the response of capital flows during periods of policy uncertainty around elections. They present evidence that foreign direct investment declines significantly prior to elections, and recovers shortly following the election. However, the response differs based on institutional quality in the country. Countries with greater institutional quality experience less variation in flows in response to policy uncertainty. Similarly, Asamoah, Adjasi and Alhassan (2016) find that macroeconomic uncertainty has a significant adverse effect on foreign direct investment flows in Sub-Saharan Africa, but that the impact on capital flows is lower in countries with stronger institutions. Country-specific levels of policy uncertainty also directly affect foreign flows. Nguyen and Lee (2021) find that countries with higher levels of policy uncertainty attract less foreign direct investment, which again is offset by more developed financial institutions. Portfolio flows are also negatively impacted by uncertainty, and monetary policy uncertainty shocks correspond to lower output, greater macroeconomic instability, and a decline in capital flows (Trung, 2022).

The financial market impact of uncertainty is also well-documented. Bordo, Duca and Koch (2016) find that economic policy uncertainty presents an inverse relationship with credit growth for financial institutions, leading to a decrease in bank credit growth by constraining bank financing. Brogaard and Detzel (2015) demonstrate the impact on uncertainty on stock market returns, indicating that it should be considered a risk factor and adequately compensated with a premium. Bahmani-Oskooee and Saha (2019) also found that policy uncertainty negatively affects stock returns for thirteen countries in the short run but not in the long run. Pastor and Veronesi (2012) argue that since investors consider uncertainty in their pricing, stocks have lower prices during periods of greater market and policy uncertainty. Ko and Lee (2015) present evidence of the significant negative effect of uncertainty on stock market returns. There is consensus that uncertainty usually has a spillover effect on other countries and the relationship between stock market returns and uncertainty is negative (Christou, Cunado, Gupta and Hassapis, 2017). However, Das and Kumar (2018) argued that the effect is lower in some emerging markets. Political uncertainty increases the risk premium and in a weaker economy, this risk premium is larger which increases the cost of borrowing (Pastor and Veronesi, 2012).

From a theoretical perspective, Manela and Moreira (2017) demonstrate the persistent effect on uncertainty on financial markets. The effects of greater uncertainty continue to impact financial markets even after controlling for forward-looking measures of volatility in stock market returns. Economic uncertainty has an adverse impact on the housing market as well. Christou, Gupta and Nyakabawo (2019) established that uncertainty negatively affects housing variables in the long run with much

significant effects on housing price, home permits and sales. Other papers have documented the negative effects of uncertainty on various asset classes, including commodities, derivatives, gold futures, bonds and stocks (Syed, Malik and Chang (2019); Hashmi and Chang (2021); Hashmi, Chang and Shahbaz (2021); Aroui, Estay and Rouband (2016); Li, Dong and Failler (2018); Fang, Chen, Yu and Qian (2018); and Uche, Chang and Gohar (2022)).

The transmission of uncertainty shocks to exchange rate conditions can come from the impact on international trade relations or from international financial flows and interest rate policies. If economic policy uncertainty spills over to policies that affect imports, exports, and trade barriers, it will in turn impact the relative demand for currency and therefore the exchange rate. Moreover, as documented in the literature above, uncertainty shocks decrease international trade flows, which may in turn affect exchange rates. Furthermore, greater global uncertainty may push risk-averse investors towards lower-risk assets in advanced economies, corresponding to net capital outflows from emerging to advanced economies. As a response to the changes in net capital flows, this would lead to greater depreciation pressure in emerging economies, and appreciation in advanced economies, which is one of the main findings of this research. Finally, uncertainty around monetary policy will influence exchange rates through the impact on interest rates. If interest rate differentials are affected by monetary policy uncertainty, then the exchange rate will also be impacted.

Kido (2016) presents evidence of the spillover of US economic policy uncertainty to exchange rates, finding a negative correlation for high-yield currencies. Others have found mixed results. Kisswani and Elian (2021) find that uncertainty impacts exchange rate volatility for some economies, but not others. Krol (2014) determines that home-country economic policy uncertainty and US economic policy uncertainty both contribute to greater exchange rate volatility during recessionary periods. However, for less open economies, home-country uncertainty has a more significant effect on exchange rates than US uncertainty. By focusing on Mexico, Bush and Noria (2021) support these findings, and demonstrate that exchange rate volatility is significantly impacted by uncertainty. The effect is, once again, amplified during recessionary periods. Abid (2020) finds that uncertainty plays a critical role in explaining both short-run and long-run currency movements, and including policy uncertainty improves the forecasting power of macroeconomic models. Beckmann and Czudaj (2017) demonstrated a new angle by considering the role of different time horizons of political decisions to study the relationship between economic policy uncertainty and exchange rate expectations. Both the degree of uncertainty regarding future policies and the announcement of the policy itself affect exchange rate expectations.

Similarly, Juhro and Phan (2018) examined how Economic Policy Uncertainty (EPU) can positively predict exchange rates and exchange rate volatility in six ASEAN countries and exchange rate volatility in ten ASEAN countries. The effect of uncertainty varies across different countries. Due to credit constraints, emerging economies are affected more by uncertainty than the US and other developed countries with more negative effects on investment and private consumption due to a shock to uncertainty (Carriere-Swallow and Cespedes, 2013). They also argued that investment and consumption are correlated with the depth of the financial markets of the respective countries. Economic policy uncer-

tainty can demonstrate an asymmetric impact on exchange rates. Using data from China's exchange rate, Liming (2020) found a positive and asymmetric effect on exchange rate volatility and that the results are dependent on the heterogeneity in different markets. However, Chang et al. (2022) studied the G7 countries and concluded that a positive shock in economic policy uncertainty significantly affects exchange rate, whereas the negative shocks do not affect exchange rates.

This paper contributes to the literature on uncertainty and exchange rates by incorporating not only exchange rate volatility, but also exchange market pressure and the real effective exchange rate, and assessing the impact of various uncertainty shocks on a global system. With greater global financial and economic integration, analyzing the impact of uncertainty shocks by also factoring in the influence of foreign and global conditions is critical.

### 3 Model & Methodology

Using the global vector autoregressive (GVAR) estimation approach is an optimal way to model the response of country-specific or region-specific factors to a global shock stemming from greater uncertainty. The model defines country-specific control variables, trade-weighted foreign variables, and global variables that shape a country's or region's response to global shocks. This research assesses the response of exchange rate conditions in emerging markets and advanced economies to global shocks stemming from heightened uncertainty. Emerging market economies are grouped into one region to analyze the response to the uncertainty shock within the context of the global system. Non-US advanced economies are also grouped into one region to analyze the response to the uncertainty shock. China, Russia and the United States are analyzed individually.

#### 3.1 Data

The analysis employs monthly data from January 1996 until December 2022 for twenty-eight countries and the euro-zone,<sup>1</sup> which covers a total of forty-eight economies and approximately 86% of world GDP. The euro-zone data is aggregated for all the countries in the region, and the bilateral trade weights are based on the region's trade with all other countries in the model, as will be explained in more detail below. The list of all countries is presented in Table 1.

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<sup>1</sup>Table 1 provides a list of all countries in the analysis.



Table 1: Countries by Region

| Advanced Economies | Emerging Markets |              | Other Economies |
|--------------------|------------------|--------------|-----------------|
| Australia          | Brazil           | Malaysia     | China           |
| Canada             | Bulgaria         | Mexico       | Russia          |
| Euro-area          | Chile            | Pakistan     | United States   |
| Japan              | Colombia         | Peru         |                 |
| Switzerland        | Czech Rep.       | Philippines  |                 |
| United Kingdom     | Hungary          | Poland       |                 |
|                    | India            | Romania      |                 |
|                    | Indonesia        | South Africa |                 |
|                    | Israel           | Thailand     |                 |
|                    | South Korea      | Turkey       |                 |

This table presents the total list of countries in the analysis. The GVAR model creates the GDP-weighted regional variables for emerging market economies. The euro-area was included using aggregated indicators for industrial production, inflation, and short-term interest rates from the OECD, as well as average regional EMPI, real effective exchange rate for the euro, and volatility for the euro.

The industrial production index, inflation based on the CPI-index, short-term domestic interest rates, and the exchange rate conditions are used for country-specific variables. The impact of global uncertainty shocks is assessed on three exchange rate conditions. First, exchange rate volatility, which is calculated as the 30-day standard deviation of the log difference of the daily local currency per US dollar exchange rate.<sup>2</sup> The measure represents the volatility of daily exchange rate movements per month. Next, the exchange market pressure index (EMPI), estimated by Patnaik, Felman and Shah (2017).<sup>3</sup> This measure represents the buildup of pressure in currency markets that must be offset by either a change in the spot exchange rate or a foreign exchange intervention by the central bank. The EMPI ( $EMP_t$ ) is calculated as:

$$EMP_t = \Delta e_t + \rho_t I_t \quad (1)$$

where  $\Delta e_t$  represents the percentage change in the local currency against the US dollar (USD),  $\rho_t$  represents the conversion factor associated with an intervention of USD 1 billion, and  $I_t$  represents the size of the foreign exchange intervention measured in billions of US dollars. The conversion factor  $\rho$  is assumed to vary over time and across countries, and is dependent on liquidity in foreign exchange markets. An increase in the exchange market pressure index represents the onset of pressure that would yield a depreciation or devaluation of the currency, whereas a decrease represents the onset of pressure that would yield an appreciation or revaluation of the currency. Finally, the real effective exchange rate, as reported by the International Monetary Fund IFS database or country-specific central bank data. An increase (decrease) in the real effective exchange rate is an appreciation (depreciation), in response to the global shock.

<sup>2</sup>For the United States, volatility is calculated as the 30-day standard deviation of the log difference of the daily US Dollar Index, as reported by FRED.

<sup>3</sup>Since the index measures exchange market pressure against the US dollar, the US EMPI is not estimated. The monthly difference in US Dollar Index is used as a proxy for exchange market pressure for the United States.

Table 2 presents the summary statistics for the country-specific variables utilized in the analysis. There is significant variability in inflation experienced across the countries, with several economies, such as Bulgaria, Romania, and Turkey, experiencing high levels of inflation in the late 1990's. Others, such as Japan and Switzerland, experienced periods of deflation through the twenty-six year time-frame. The industrial production index (IPI) also varies greatly across the countries in the analysis, with the lowest industrial production experienced in Poland in January 1996, of 28.89. The highest of 267.2 was observed in China in June 2022. The interest rates have significant variance across time and countries, with the highest interest rate observed in Turkey in January 1996, of 204.4%, while the lowest rates found in Switzerland in August 2019, of -0.86%.

Table 2: Country Variable Summary Statistics

|          | Inflation | IPI   | Interest Rates | FX Volatility | EMPI   | RER   |
|----------|-----------|-------|----------------|---------------|--------|-------|
| Max      | 177.4     | 267.2 | 204.4          | 93.14         | 121.1  | 156.2 |
| Min      | -4.40     | 28.89 | -0.86          | 0.00          | -56.97 | 34.47 |
| Average  | 5.70      | 94.50 | 6.98           | 0.73          | 0.26   | 97.38 |
| Median   | 3.20      | 97.30 | 4.22           | 0.48          | 0.10   | 97.36 |
| Variance | 126.6     | 666.3 | 142.31         | 3.78          | 27.82  | 320.5 |
| Kurtosis | 70.70     | 3.90  | 59.27          | 103.2         | 44.07  | 6.25  |

This table presents the summary statistics for the key country-specific variables used in the analysis. The data is summarized across all thirty countries in the data set.

The exchange rate conditions vary across countries, with the highest level of volatility experienced by Indonesia in January 1996, while the lowest in several countries in different time periods. Exchange market pressure also varies across countries. The highest EMPI was in Bulgaria in February 1997, while the lowest in Switzerland in July 2011.

There are five measures of uncertainty that are assessed. The first is the Global Economic Policy Uncertainty Index, calculated by Baker et al. (2016), measures the GDP-weighted average of national economic policy uncertainty indices for 20 countries: Australia, Brazil, Canada, Chile, China, France, Germany, Greece, India, Ireland, Italy, Japan, Mexico, the Netherlands, Russia, South Korea, Spain, Sweden, the United Kingdom, and the United States. This measure of uncertainty provides a global perspective on the effects of heightened uncertainty across the largest economies in the world, and how that may spillover to other economies, particularly in this research, emerging markets.

Next, the US monetary policy uncertainty index represents the measure of uncertainty based on news articles and reports regarding US monetary policy and Federal Reserve action (Baker et al., 2016). This measure allows us to focus on the transmission of shocks stemming from the US monetary policy to emerging markets, which is critical in understanding the impact on exchange rate dynamics for these economies. Most emerging markets have a significant portion of foreign exchange reserves held in US dollars, and intervene in currency markets to manage their currency values, smooth volatility, or increase liquidity by buying and selling US dollars. Therefore, uncertainty around US monetary policy action may have a significant effect on the dynamics of exchange rate conditions for emerging markets.

Third, the equity market volatility index for the US serves as a proxy for financial market uncertainty, and is reported by the Federal Reserve's FRED database. Heightened volatility corresponds to greater financial stress and uncertainty, and may reflect underlying uncertainty around business conditions. The focus is on volatility in US equity markets due to the size and significance of this market for foreign investors.

The fourth measure of uncertainty, estimated by Puttman (2018), is the financial stress indicator. This measure tracks references of financial stress in news articles from the United States regarding business conditions, central bank policy action, asset and gold prices, as well as inflation and stock prices to compile an index based on the negative connotations associated with the terms in the articles. The final measure of uncertainty is estimated by Caldara and Iacoviello (2022) and tracks geopolitical risk on a global scale. Similar to the other indices of uncertainty, this index tracks the number of news articles that are related to adverse geopolitical events and rising geopolitical tensions, focusing on threats of war, nuclear threats, military buildup, terror threats, and escalation of aggression between countries.

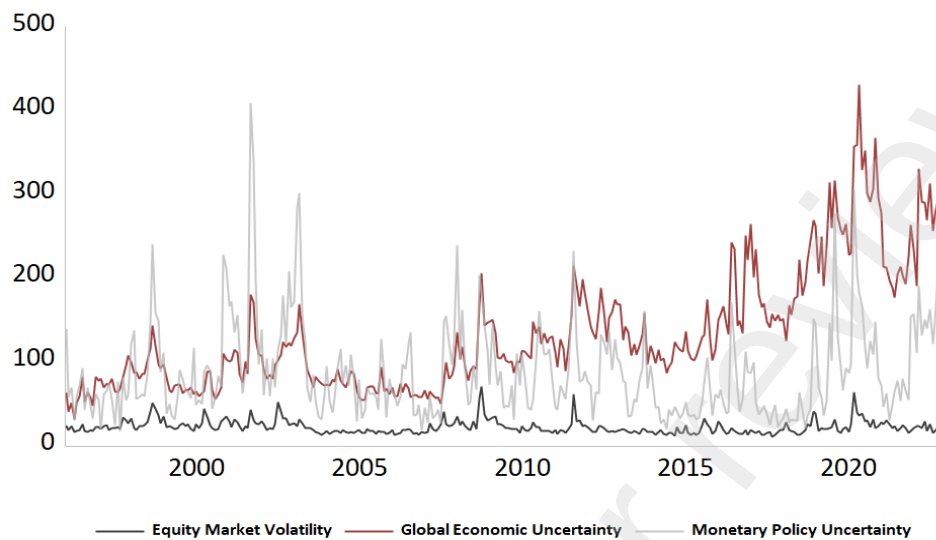
Table 3: Global Uncertainty Variable Summary Statistics

|          | Global Economic<br>Uncertainty | Monetary Policy<br>Uncertainty | Equity Market<br>Volatility | Financial Stress<br>Index | Geopolitical Risk<br>Index |
|----------|--------------------------------|--------------------------------|-----------------------------|---------------------------|----------------------------|
| Max      | 430.2                          | 407.9                          | 69.84                       | 104.50                    | 512.53                     |
| Min      | 30.69                          | 16.57                          | 9.57                        | 99.88                     | 39.05                      |
| Average  | 132.7                          | 87.38                          | 21.39                       | 101.22                    | 99.33                      |
| Median   | 110.5                          | 71.13                          | 19.41                       | 101.22                    | 88.66                      |
| Variance | 5,322.3                        | 3,224.5                        | 68.16                       | 0.80                      | 51.07                      |
| Kurtosis | 1.34                           | 5.03                           | 7.89                        | 1.12                      | 27.59                      |

This table presents the summary statistics for the global variables used in the analysis to measure uncertainty.

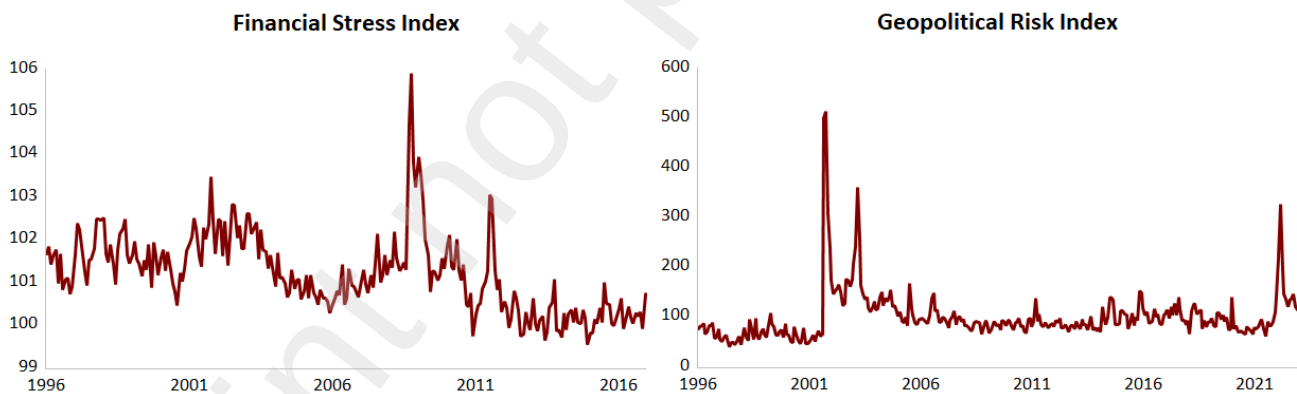
With the duration of 26 years of data, including several recessions, the Global Financial Crisis, the COVID-19 pandemic, and the Russian invasion of Ukraine, the model analyzes periods of substantial buildup of uncertainty around the world. Table 3, presents the summary statistics for the five measures of uncertainty. Most notably, the variance in uncertainty is notably high across several of the measures, and in particular for global economic and US monetary policy uncertainty. As can be seen from Figure 1, global economic uncertainty was highest from April to June of 2022, corresponding to the spread of Covid-19 and the impact of lock-downs on economic performance. US monetary policy uncertainty was highest in September 2001, following the 9/11 attacks on the World Trade Center, when uncertainty spiked to 407.9. Other periods of heightened uncertainty include March 2020 (304.7), corresponding to the onset of Covid-19 in the US, and January 2008 (238.3) corresponding to the further deterioration of the US economy during the Global Financial Crisis. Equity market volatility was highest in October 2008, during the peak of the sub-prime mortgage crisis, as well as during periods in August 2011, corresponding to the fear of contagion from the European Debt Crisis, and March 2020, with the onset of lockdown measures to prevent the spread of Covid-19.

Figure 1: Global Uncertainty Measures



This figure depicts the Global Economic Uncertainty index, the US Monetary Policy Uncertainty index, and the Equity Market Volatility index from January 1996 to December 2022.

Figure 2: Global Uncertainty Measures



This figure depicts the Financial Stress Index and the Geopolitical Risk Index, estimated by Puttman (2018) and Caldara and Iacoviello (2022) respectively.

Figure 2 presents the Financial Stress Index and the Geopolitical Risk Index. The Financial Stress Index (FSI) is estimated until December 2016, however it provides a strong depiction of periods of heightened financial uncertainty. It showcases the significant spike in uncertainty around the Global Financial Crisis, beginning in 2007 through 2009. Additionally, the increase in the index in 2011 and 2012 corresponds to the European Debt Crisis, and concern about the collapse of the euro. The

Geopolitical Risk Index (GPR) presents uncertainty around global security and geopolitical tensions. This indicator identifies high levels of uncertainty in 2001, as a result of the September 11th terrorist attacks, as well as the 2022 Russian invasion of Ukraine.

### 3.2 Empirical Model

The GVAR model estimates country-specific VAR models, followed by the global VAR model, and then introduces shocks to measure the impulse response functions. The first stage of the GVAR model determines stationarity and cointegration among the domestic, foreign, and global variables. The model then estimates each country's vector autoregression (VARX) model which consists of a set of endogenous domestic variables,  $x_{it}$ , a set of weakly exogenous foreign variables,  $x_{it}^*$ , and weakly exogenous global variables,  $g_t$ . The foreign variables are estimated as:

$$x_{it}^* = \sum_{j=1}^N w_{ij} x_{jt} \quad (2)$$

where  $w_{ij}$  represent the trade-based weights between the estimated country and all other countries in the system, determined by the bilateral trade flows. The trade-weighted matrix plays a role in creating the linkages between countries and across different regions. It represents the degree to which one country or region depends on all other countries or regions in the system (Pesaran, Schuermann and Weiner, 2004). The relationship for each country with the foreign variables serves as a proxy for global unobserved factors that may impact the response to the global shock.

In the first step of the country-specific estimation, the VARX( $p, q, s$ ) of country  $i$ , with  $p$  lagged order of  $x_{it}$ ,  $q$  lagged order of  $x_{it}^*$ , and  $s$  lagged order of  $g_t$  can be written as:

$$x_{it} = \alpha_{i0} + \alpha_{i1t} + \sum_{j=1}^p \phi_{ij} x_{i,t-j} + \sum_{k=1}^q \lambda_{ik} x_{i,t-k}^* + \sum_{l=1}^s \nu_{il} g_{i,t-l} + \mu_{it} \quad (3)$$

where the lagged orders of  $p, q, s$  are selected based on the Akaike Information Criterion (AIC).  $x_{it}$  is the  $k_i \times 1$  vector of domestic variables of country  $i$ ,  $x_{it}^*$  is the  $k_i^* \times 1$  vector of weakly exogenous foreign variables of country  $i$ , and  $g_t$  is the  $k_g \times 1$  vector of weakly exogenous global commonly observed variables. The error term,  $\mu_{it}$ , is independent and identically distributed (i.i.d.) within the country specific VARX equation. Across the country models in the GVAR, it is assumed to be weakly dependent. This enables the transmission of idiosyncratic shocks from an individual country model to all other countries in the system.

Assuming integration of order one, the country-specific VARX allows for cointegration among the domestic, foreign, and global variables. As an example, the VARX(1,1,1) can be re-parameterized into the error-correction form VECMX(1,1,1) with rank order  $r_i$ :

$$\Delta x_{it} = c_{i0} - \alpha_i \beta_i' [z_{i,t-1} - \gamma_{i,t-1}] + \lambda_{i0} \Delta x_{it}^* + \nu_{i0} \Delta g_{it} + \mu_{it} \quad (4)$$

where  $z_{it} = (x'_{it}, x'^{*}_{it}, g'_t)'$ ,  $\beta_i$  is a  $(k_i + k_i^* + k_g) \times r_i$  matrix of cointegration coefficients,  $\alpha_i$  is a  $k_i \times r_i$  matrix that captures the speed of the adjustment. By defining  $\beta_i = (\beta'_{ix}, \beta'_{ix^*}, \beta'_{ig})$ , the error correction term can be written as:

$$\beta'_i[z_{i,t-1} - \gamma_{i,t-1}] = \beta'_{ix}x_{it} + \beta'_{ix^*}x^*_{it} + \beta'_{ig}g_t - (\beta'_i\gamma_i)t \quad (5)$$

In the estimation, the domestic variables for each country are the exchange rate condition, which is either exchange rate volatility, exchange market pressure, or the real effective exchange rate depending on the model, the domestic interest rate, domestic industrial production index, and the inflation rate. The foreign variables include interest rates, industrial production index, and inflation for other countries in the system. Based on the Augmented Dickey-Fuller test, the variables require monthly first differencing to ensure stationarity.<sup>4</sup>

The global variables are the price of oil and the uncertainty measure. The model employs five different measures of uncertainty: Global Economic Policy Uncertainty, US Monetary Policy Uncertainty, US Equity Market Volatility, Financial Stress Index, and Geopolitical Risk Index. However, we run each model with only one uncertainty measure at a time. Therefore, we estimate fifteen models, five separate uncertainty shocks per three exchange rate conditions. All models employ data from 1996 to 2022 on a monthly basis, except for the Financial Stress Index, which ends in 2016. For the Financial Stress Index shock, the model is run with monthly data from 1996 to 2016 for all variables.

The second step of the GVAR model, after estimating the country-specific VARX and VECMX models, is to stack the individual country VARX equations and generate a global model. Since  $z_{it} = (x'_{it}, x'^{*}_{it}, g'_t)'$ , then  $x_t = W_i^{-1}z_{it}$ , where the weight matrix  $W_i$  is a  $(k_i + k_i^* + k_g) \times k$  matrix and  $x_t = (x'_{1t}, \dots, x'_{Nt})'$  is a  $k \times 1$  vector that includes all variables. Assuming  $p = q = s$ , the GVAR (p) model is then:

$$x_t = G_0^{-1}\alpha_0 + G_0^{-1}\alpha_{1t} + G_0^{-1}G_1x_{t-1} + \dots + G_0^{-1}G_px_{t-p} + G_0^{-1}\mu_t \quad (6)$$

where  $\alpha_0, \alpha_1, G_0, G_1, \dots, G_p$  are all country-specific vectors. All of the variables in the model are then solved simultaneously. The GVAR estimation constructs impulse response functions to present the response of variables in individual countries or regions to a shock that affects the whole system. The global shock is introduced as a one standard deviation shock in each of the five uncertainty measures presented above. The generalized impulse response functions are estimated, so that the order of the factors does not influence the results of the responses to the shocks. Further details of the GVAR model are presented in Chudik and Pesaran (2016), Dees, di Mauro, Pesaran and Smith (2007) and Pesaran et al. (2004).

The GVAR model allows for the grouping of countries into regions, thereby estimating first the country-specific VARX model for the region. The regional weights are determined by the sum of GDP (in purchasing power parity terms) and each country's share of the total regional GDP. The GDP-weights

<sup>4</sup>With the exception of exchange rate volatility.

for the region add up to one, with each country assigned a unique weight based on their share of the overall regional GDP. The regional-GDP weights are used for computing regional variables, shocks, and aggregation of impulse response functions. Once the GVAR is estimated for the region, the impulse response functions are then presented for the region, rather than the country. The regional weights are not the same as the bilateral trade-weights used in the model estimation for foreign variables; rather they determine the composition of the responses based on the countries within the given region.

Twenty countries from the global model are grouped in the emerging markets region, and China and Russia are analyzed individually. The emerging markets are grouped into one region specifically to aggregate the results and determine the impact of uncertainty shocks on exchange rate conditions for several countries at once, with similar characteristics as semi-open, fast-growing economies, that may not have fully developed financial markets and institutions. China is analyzed separately from the other emerging markets due to its size, both in GDP terms and share of trade with each country, thereby distorting the emerging market results to reflect, mainly, the response of the China. Russia is analyzed separately as well due to its dependence on oil as a primary export and source of foreign currency. Therefore, it is interesting to assess Russia as a unique case among the emerging markets.

The non-US advanced economies are grouped together, which include five advanced economies (Australia, Canada, Japan, Switzerland and the United Kingdom) and the Euro-zone. The US is analyzed individually, due to the dominance of the US dollar on global financial markets and the international transmission of policy shocks stemming from the US to the rest of the world. The countries included in each region are presented in the Appendix.

## 4 Empirical Results

This section presents the results of the GVAR model by focusing on the impact of uncertainty shocks on exchange rate measures for non-US advanced economies, emerging market economies, China, Russia and the United States. The segmentation of groups matters for the consideration of the impact of global uncertainty shocks for core currencies (advanced economies and US) versus periphery currencies (emerging), as well as aspiring global economic powers such as China and, to a lesser degree, Russia. The segmentation of the impulse response functions based on these regions illuminates whether there are differences in responsiveness of exchange market conditions to heightened periods of uncertainty.

The GVAR model is estimated by following the methodology of Smith and Galesi (2014), and our model is first estimated in the error correction form if there exists cointegration among the variables. The estimation requires the underlying variables to be approximately integrated of order one. Therefore, unit root tests are employed to ensure stationarity of the variables. We present the results for the standard Augmented Dickey-Fuller tests, with and without a trend, in Table 4. All variables, except exchange rate volatility, are represented in the model as monthly first differences. As seen below, all variables in the model are stationary.

Table 4: Unit Root Tests for GVAR Variables

|                                  | Emerging | China  | Russia | Advanced | US     | Global |
|----------------------------------|----------|--------|--------|----------|--------|--------|
| Exchange Rate Volatility (Trend) | -6.60    | -8.38  | -6.91  | -4.99    | -5.90  |        |
| (No trend)                       | -6.60    | -8.47  | -7.03  | -4.89    | -5.84  |        |
| EMPI                             | -7.20    | -4.52  | -6.50  | -6.65    | -11.84 |        |
|                                  | -7.21    | -4.34  | -6.51  | -6.57    | -11.86 |        |
| Real Effective Rates             | -12.10   | -11.49 | -11.05 | -9.35    | -11.73 |        |
|                                  | -12.12   | -11.51 | -11.01 | -9.29    | -11.67 |        |
| IPI                              | -14.86   | -17.07 | -20.70 | -15.26   | -10.93 |        |
|                                  | -14.88   | -16.92 | -20.72 | -15.35   | -10.89 |        |
| Interest Rates                   | -10.42   | -9.57  | -10.49 | -4.95    | -4.82  |        |
|                                  | -10.31   | -9.30  | -10.30 | -5.10    | -4.70  |        |
| Inflation                        | -8.72    | -10.98 | -6.75  | -10.14   | -11.63 |        |
|                                  | -8.69    | -11.00 | -5.93  | -10.20   | -11.72 |        |
| Global Economic Uncertainty      |          |        |        |          |        | -10.91 |
|                                  |          |        |        |          |        | -10.92 |
| US Monetary Policy Uncertainty   |          |        |        |          |        | -11.05 |
|                                  |          |        |        |          |        | -11.08 |
| Equity Market Volatility         |          |        |        |          |        | -11.65 |
|                                  |          |        |        |          |        | -11.67 |
| Financial Stress Indicator       |          |        |        |          |        | -9.98  |
|                                  |          |        |        |          |        | -10.00 |
| Geopolitical Risk Index          |          |        |        |          |        | -11.25 |
|                                  |          |        |        |          |        | -11.26 |

The table presents the values for the Augmented Dickey-Fuller unit root test statistic, with (without) trend, for the regions and variables used in the GVAR model. The critical value for all variables is -3.45 (-2.89).

Next, Table 5 presents the F-statistics for the weak exogeneity tests for each regional model. The F-critical values are at the 5% significance level, and the null hypothesis is that weak exogeneity exists. Weak exogeneity demonstrates a unidirectional long-run force from the foreign/global variable to the domestic variable, while also allowing for lagged mutual effects between them in the short run. The model assumes idiosyncratic shocks in the country-specific and region-specific estimation should be weakly correlated, thereby establishing the appropriateness of estimating the transmission of shocks within the context of the global vector autoregression. In our estimation, only the industrial production index statistics reject the null hypothesis of weak exogeneity in all non-US economies. Therefore, industrial production changes may not yield a significant effect on the response to the global shocks. The remaining statistics do not reject the null hypothesis, and we can assume presence of weak exogeneity among the foreign/global variables with the domestic variables in the remaining regional models. Table 5 also presents the number of cointegrating relationships for each region-specific model. All regions and model variations demonstrate at least one cointegrating relationship.



Table 5: Weak Exogeneity Tests for GVAR Model

|                                | F-Critical Value | Inflation Rate | IPI    | Interest Rate | Global Condition | # Co-int. Relations |
|--------------------------------|------------------|----------------|--------|---------------|------------------|---------------------|
| Emerging Market Economies      |                  |                |        |               |                  |                     |
| Global Economic Uncertainty    | 3.03             | 0.26           | 21.39* | 0.46          | 1.19             | 2                   |
| US Monetary Policy Uncertainty | 3.03             | 0.64           | 21.80* | 0.39          | 1.24             | 2                   |
| Equity Market Volatility       | 3.03             | 0.55           | 20.46* | 0.38          | 0.72             | 2                   |
| Financial Stress Indicator     | 3.03             | 0.96           | 0.82   | 2.76          | 0.24             | 2                   |
| Geopolitical Risk Index        | 3.03             | 0.19           | 24.25* | 0.46          | 0.46             | 2                   |
| China                          |                  |                |        |               |                  |                     |
| Global Economic Uncertainty    | 3.03             | 0.67           | 4.56*  | 1.05          | 0.67             | 2                   |
| US Monetary Policy Uncertainty | 3.03             | 0.54           | 4.31*  | 1.00          | 0.05             | 2                   |
| Equity Market Volatility       | 3.03             | 0.61           | 4.21*  | 0.99          | 0.60             | 2                   |
| Financial Stress Indicator     | 3.03             | 0.14           | 0.70   | 1.15          | 0.41             | 2                   |
| Geopolitical Risk Index        | 3.03             | 0.60           | 3.99*  | 0.99          | 0.17             | 2                   |
| Russia                         |                  |                |        |               |                  |                     |
| Global Economic Uncertainty    | 3.03             | 0.30           | 5.70*  | 3.34          | 0.80             | 2                   |
| US Monetary Policy Uncertainty | 3.03             | 0.20           | 5.66*  | 3.41          | 0.37             | 2                   |
| Equity Market Volatility       | 3.03             | 0.22           | 4.53*  | 3.28          | 0.36             | 2                   |
| Financial Stress Indicator     | 3.03             | 0.48           | 3.06*  | 1.55          | 1.58             | 2                   |
| Geopolitical Risk Index        | 3.03             | 0.32           | 5.79*  | 3.49          | 0.01             | 2                   |
| Non-US Advanced Economies      |                  |                |        |               |                  |                     |
| Global Economic Uncertainty    | 3.87             | 0.09           | 26.97* | 0.12          | 0.45             | 1                   |
| US Monetary Policy Uncertainty | 3.87             | 0.00           | 32.19* | 0.15          | 1.29             | 1                   |
| Equity Market Volatility       | 3.87             | 0.01           | 29.00* | 0.04          | 0.20             | 1                   |
| Financial Stress Indicator     | 3.88             | 0.42           | 5.57*  | 0.16          | 0.24             | 1                   |
| Geopolitical Risk Index        | 3.87             | 0.00           | 32.09* | 0.14          | 0.07             | 1                   |
| United States                  |                  |                |        |               |                  |                     |
| Global Economic Uncertainty    | 3.87             | 0.01           | 2.03   | 0.24          | 0.68             | 1                   |
| US Monetary Policy Uncertainty | 3.87             | 0.00           | 2.99   | 0.09          | 0.88             | 1                   |
| Equity Market Volatility       | 3.87             | 0.07           | 2.50   | 0.16          | 0.57             | 1                   |
| Financial Stress Indicator     | 3.88             | 0.01           | 0.43   | 0.01          | 0.08             | 1                   |
| Geopolitical Risk Index        | 3.87             | 0.01           | 2.96   | 0.13          | 0.13             | 1                   |

The table presents the F-statistics for the weak exogeneity tests for the foreign and global variables used in the regional GVAR model estimations. The F-test critical values are for the 5% level significance. \* represents statistical significance, and rejection of the null hypothesis of weak exogeneity. The final column presents the number of cointegrating relationships detected in the VECMX estimation.

The emerging market economies together are grouped together, and China and Russia are treated separately in the estimation, as explained in Section 3.2. Figure 3 presents the response of exchange rate volatility to a one standard deviation shock in the uncertainty measures. Overall, all five measures of uncertainty correspond to an increase in exchange rate volatility for emerging market economies. For China, although exchange rate volatility does increase in response to heightened uncertainty, the response of volatility to the shock is much smaller. This may be due to the exchange rate management and interventions that are regularly utilized by the People's Bank of China, and therefore not surprising, since one of the goals of foreign exchange intervention is to smooth excessive volatility. Shocks in

geopolitical risk and uncertainty affect emerging markets with a lag, leading to an increase in volatility one or two periods after the shock. However, the geopolitical uncertainty shock does not have a statistically significant impact on exchange rate volatility. Russia's exchange rate volatility responds to shocks in global economic policy uncertainty, US monetary policy uncertainty and equity market volatility. However, the response is not significant to shocks in financial stress, nor for geopolitical risk.

Figure 4 presents the response of exchange market pressure to shocks in global uncertainty measures. This measures represents the buildup of pressure in currency markets that must be offset by an appreciation/depreciation or foreign exchange intervention. Across emerging market economies, an uncertainty shock corresponds with an immediate increase in exchange market pressure. The positive response of the index to a one standard deviation shock in global economic uncertainty, US monetary policy uncertainty, equity market volatility, and financial stress implies that emerging market currencies experience an buildup of depreciationary pressure as a result of these shocks. It is a transient response, that quickly stabilizes and becomes not significant after one period. The response is not significant for shocks in geopolitical uncertainty and risk.

The Russian EMPI mirrors the response in other emerging market economies, with a significant increase in exchange market pressure immediately following a one standard deviation shock in global economic uncertainty, US monetary policy uncertainty, and equity market volatility. The response in Russian EMPI is larger in magnitude compared to other emerging market economies and remains persistently higher after the shock. Additionally, there is a significant and persistent increase in exchange market pressure in Russia to a shock in the financial stress index. Overall, the buildup of pressure for Russia in currency markets after an uncertainty shock is more sizable and persistent than for other emerging market economies. Again, geopolitical risk does not impact exchange market pressure significantly. For China, the only significant response of the EMPI is to a shock in global economic uncertainty, while the other measures do not have a significant impact on China's exchange market pressure. A one standard deviation shock in global economic uncertainty corresponds to an immediate increase in exchange market pressure, however it is smaller in magnitude than for other emerging markets and quickly dissipates. The response to financial stress in the United States appears to lead to a persistent increase in exchange market pressure for China.

Figure 5 presents the response of real effective exchange rates to shocks in global uncertainty in emerging markets. Overall, real effective exchange rates depreciate sizably and significantly immediately following a one standard deviation shock in global economic uncertainty, US monetary policy uncertainty, equity market volatility and financial stress. The depreciation recovers after one to two periods, and dissipates. The response is largest in magnitude in Russia and smallest in magnitude in China. The response to financial stress is not significant for these two large emerging markets, whereas the response in Russia to geopolitical risk corresponds to a significant depreciation.

Overall, it is clear that exchange market conditions across emerging market economies respond significantly to shocks in global economic uncertainty. There is an immediate increase in volatility, a buildup of exchange market pressure, and a depreciation in the real effective exchange rate when

there is heightened global uncertainty. The response is most clear and significant to shocks in global economic uncertainty, US monetary policy uncertainty, and equity market volatility. Moreover, the response of these conditions to the shocks is occurring within the context of the global system in the GVAR model. Therefore, the impulse response functions take into account the economic conditions of both the domestic and foreign countries, as well as global factors that may influence the response to global shocks. The period of analysis covers several episodes of significant global uncertainty, and therefore is not limited just to the recent periods corresponding to the global health pandemic and geopolitical turmoil. It is possible that the more significant response by Russia's exchange market conditions correspond to the impact of the Ukrainian invasion, and the global response in opposition to the invasion, on Russia's economy and the value of the ruble.

Next, the results are presented for the advanced economies' exchange market conditions response to heightened global economic uncertainty, which also represents the response of global "core currencies" to spikes in uncertainty. Core currencies those which are used most frequently to settle foreign exchange transactions and in international banks, including the US dollar, Euro, Japanese Yen, and British Pound Sterling. The grouping of non-US advanced economies also includes Australia, Canada and Switzerland.

Figure 6 presents the response of exchange rate volatility to a spike in global uncertainty. Similar to the emerging market economies, exchange rate volatility increases sizably and significantly in advanced economies in response to a shock in global economic uncertainty, US monetary policy uncertainty, and equity market volatility. It is an immediate response to the shock in uncertainty, and dissipates after one to two periods. The response is smaller in magnitude and significance to a shock in geopolitical uncertainty.

Figure 7 presents the response of exchange market pressure to shocks in uncertainty in the advanced economies. Whereas emerging markets' EMPI responds to the shock with a significant increase, denoting significant depreciation pressure, the opposite response occurs for advanced economies. Specifically, shocks in global economic, monetary, and financial uncertainty correspond to immediate buildup of appreciation pressures. In the United States, despite a small, immediate decrease, the dollar index increases after one period and then corrects.<sup>5</sup> The US dollar also responds immediately to a shock in geopolitical risk, corresponding to an increase in the index.

Finally, Figure 8 presents the response of the real effective exchange rate to shocks in uncertainty in the advanced economies. Supporting the findings from Figure 7, the impulse response functions indicate that spikes in global economic, monetary, and financial uncertainty correspond to a sizable, significant and immediate appreciation of the real effective exchange rate in advanced economies. This is followed by a slight "correction" after one period, with a depreciation that then dissipates after one to two periods. The response is significant for all measures except geopolitical risk. Geopolitical uncertainty, overall, does not demonstrate a significant or strong effect on exchange rate conditions in either advanced or emerging economies. However, in the United States, it does correspond to an increase in volatility, an increase in the US Dollar Index, and depreciation of the real effective exchange

<sup>5</sup>Recall that the US EMPI is proxied by the US Dollar Index, since the EMPI is not available for the United States.

rate, albeit temporary. As an extension of the research, it will be useful to assess in the country-specific uncertainty shocks related to geopolitical impact exchange rate conditions. In the current estimation, the major events include the September 11th attacks in 2001, the 2003 invasion of Iraq, and the 2022 Russian invasion of Ukraine. To understand country-level uncertainty indices may provide additional insight on the relative effect of geopolitical events on foreign exchange conditions.

## 5 Conclusion

In an increasingly globalized and interconnected world, there is a growing significance in examining the implications of economic and policy uncertainty. Over the past few decades, we have witnessed notable events such as the Global Financial Crisis, the European Debt Crisis, the Covid-19 pandemic, the Russian invasion of Ukraine and several other incidents. These recent historical events have led to a significant increase in global economic uncertainty, corresponding to a renewed interest in the academic literature to studying the macroeconomic effects of uncertainty shocks.

Uncertainty can come in multiple forms through policy changes, geopolitical turmoil, and economic or financial instability. It affects economic growth, consumption, investment, employment and international flow of goods and services negatively. The research presented in this paper contributes to the literature by focusing on the response of exchange market conditions to spikes in global economic policy uncertainty, monetary policy uncertainty, financial uncertainty, and geopolitical uncertainty.

Overall, significant increases in uncertainty measures lead to an immediate increase in exchange rate volatility across emerging and advanced economies. The shocks also spillover to affect exchange market pressure and real effective exchange rates, although the direction of the effect depends on whether the currencies are core versus periphery currencies. In emerging markets, with periphery currencies, uncertainty shocks correspond to greater depreciation in real effective exchange rates as well as a buildup of depreciation pressure in the exchange market pressure index. The effect is most pronounced for shocks in global economic uncertainty, US monetary policy uncertainty, and equity market volatility. However, China exhibits a comparatively subdued response, which aligns with China's active interventions and control over the value of the Chinese Yuan. In contrast, Russia experiences a more pronounced and significant reaction compared to the other emerging markets.

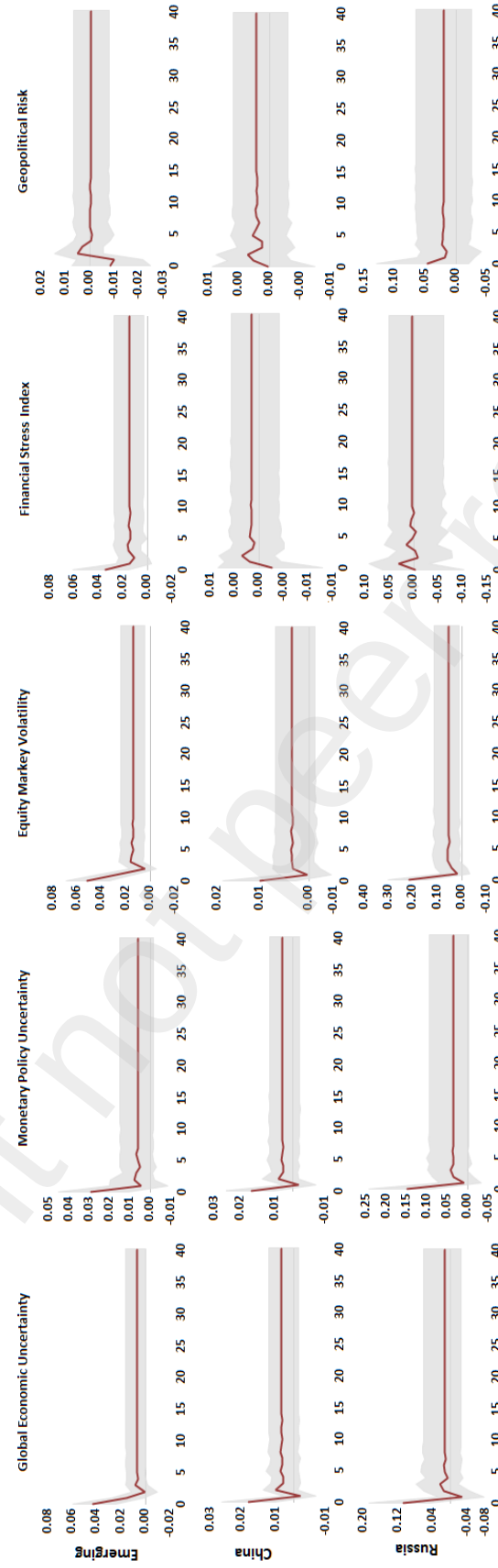
In advanced economies, corresponding to core currencies, shocks in uncertainty lead to appreciation in the real effective exchange rate and a buildup of appreciation in the exchange market pressure index. The difference between emerging and advanced economies in their exchange rate responses may be linked to the effects on international financial flows. Greater uncertainty may motivate foreign investors to move towards less risky assets in advanced economies. As they respond to uncertain global economic and financial conditions in the adjustment of their portfolios, this corresponds to the buildup of appreciate/depreciation pressures in advanced/emerging markets, respectively.

Furthermore, by assessing the response of exchange market conditions to uncertainty shocks in the global vector autoregression model, a global system of countries is created. The response of exchange

rate conditions in one country to an uncertainty shock controls for the country-specific variables, trade-weighted foreign variables, and global variables that may influence the response to the shock. Therefore, this model provides a holistic analysis of the exchange rate response to uncertainty shocks within the framework of a global system. By including twenty-eight countries and the euro-zone, the global system represents 86% of the world economy, in terms of GDP.

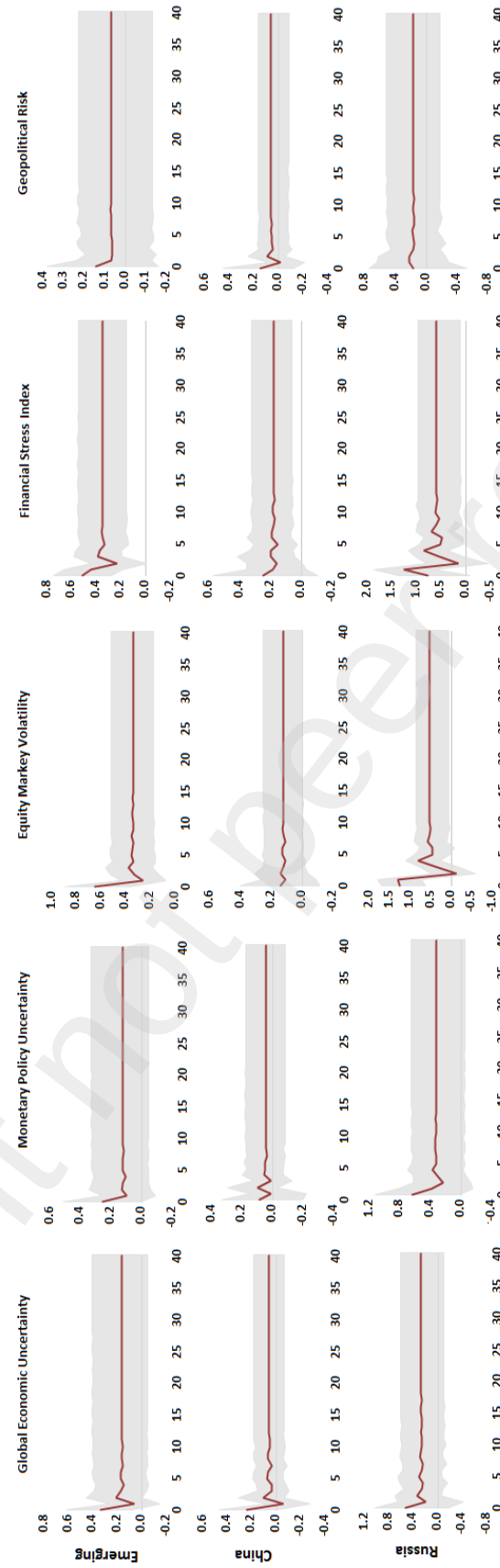
The findings of this research yield important policy implications, especially for vulnerable emerging market economies. First, the spillover of uncertainty shocks from advanced economies, especially from the United States, have significant effects on domestic currencies and exchange rate volatility. In periods of heightened uncertainty, policymakers in emerging markets may require foreign exchange interventions to buffer against macroeconomic effects of uncertainty shocks, even when those shocks do not originate in their country. Second, the transmission of shocks may be most significant through financial markets and financial flows. While formulating policies, the timing of policy implementation and the risks that might arise due to interdependencies among the countries should be taken into consideration. Policies that create buffers or incentives to alleviate the risk-adjustment response of foreign investors may be required. Finally, the level of financial depth, development, and integration into global markets may have a notable effect on the size of the shock response and is an area that requires further research.

Figure 3: Response of Exchange Rate Volatility to Uncertainty Shocks in Emerging Markets



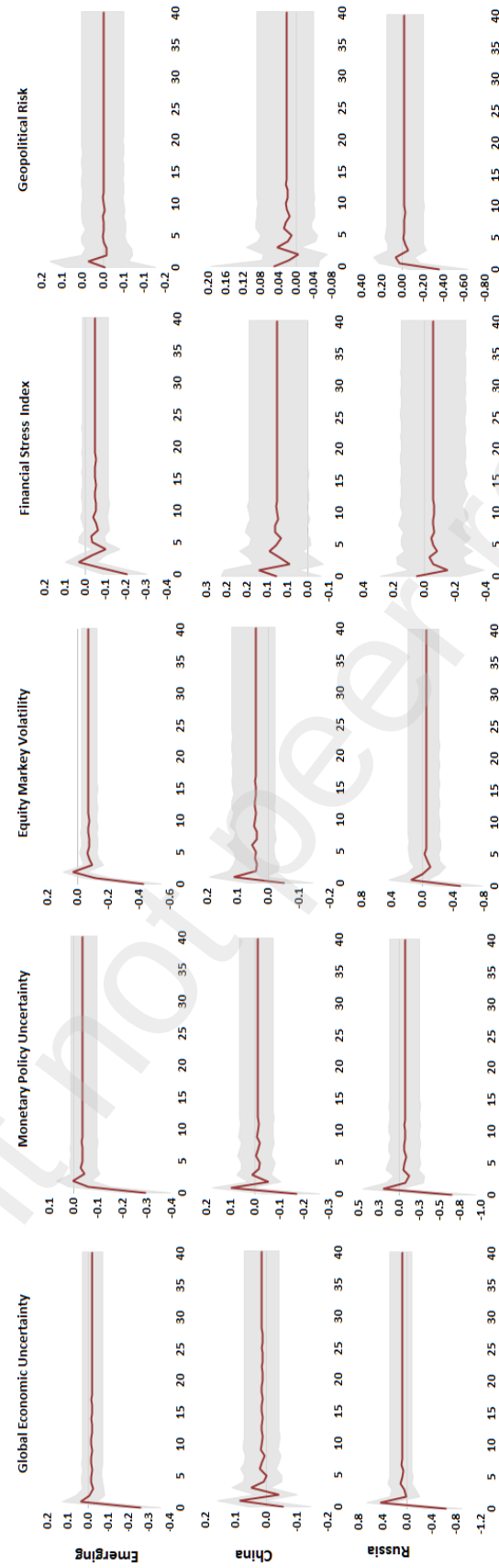
This figure depicts the response of exchange rate volatility in emerging markets, China and Russia to a shock in global uncertainty, measured by five different indicators. The shaded gray areas represent the 95% confidence intervals.

Figure 4: Response of Exchange Market Pressure to Uncertainty Shocks in Emerging Markets



This figure depicts the response of exchange market pressure in emerging markets, China and Russia to a shock in global uncertainty, measured by five different indicators. The shaded gray areas represent the 95% confidence intervals.

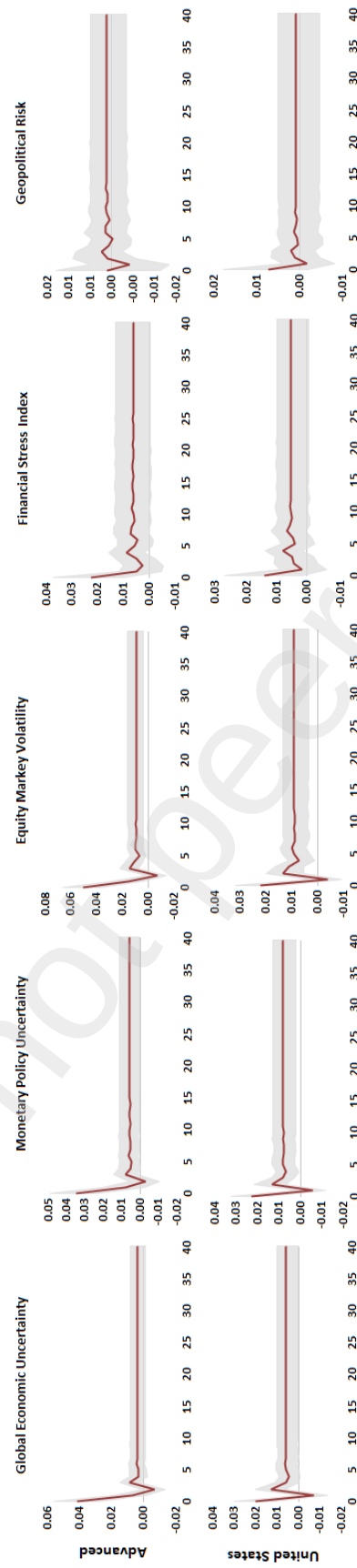
Figure 5: Response of the Real Effective Exchange Rate to Uncertainty Shocks in Emerging Markets



This figure depicts the response of the real effective exchange rate in emerging markets, China and Russia to a shock in global uncertainty, measured by five different indicators. The shaded gray areas represent the 95% confidence intervals.

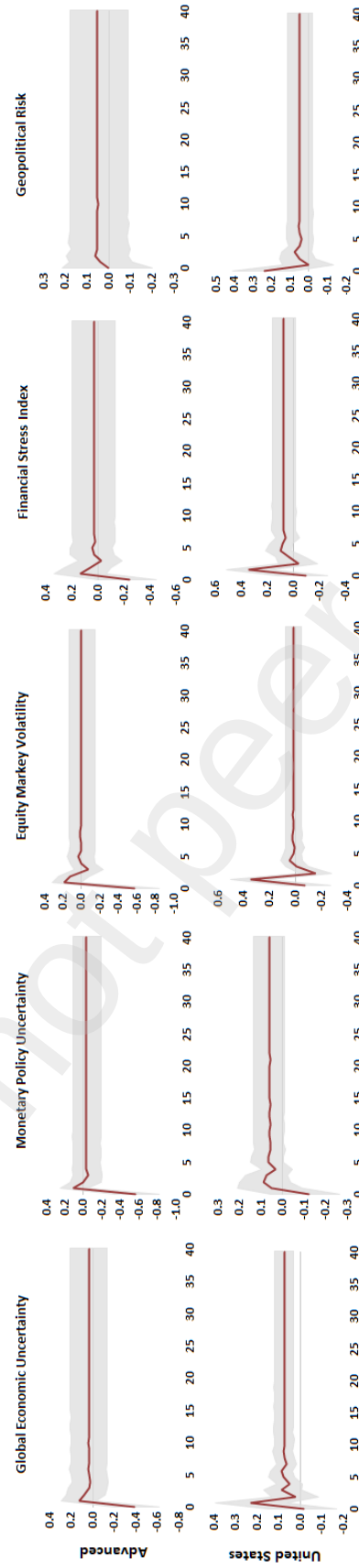


Figure 6: Response of Exchange Rate Volatility to Uncertainty Shocks in Advanced Economies



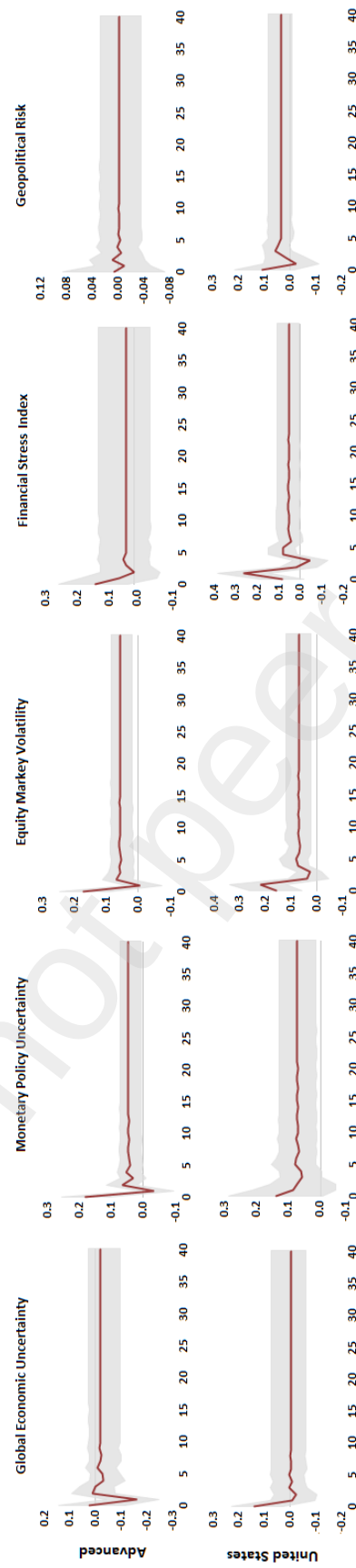
This figure depicts the response of exchange rate volatility in five advanced economies to a shock in global uncertainty, measured by five different indicators. The shaded gray areas represent the 95% confidence intervals.

Figure 7: Response of Exchange Market Pressure to Uncertainty Shocks in Advanced Economies



This figure depicts the response of exchange market pressure in five advanced economies to a shock in global uncertainty, measured by five different indicators. The shaded gray areas represent the 95% confidence intervals.

Figure 8: Response of the Real Effective Exchange Rate to Uncertainty Shocks in Advanced Economies



This figure depicts the response of the real effective exchange rate in five advanced economies to a shock in global uncertainty, measured by five different indicators. The shaded gray areas represent the 95% confidence intervals.

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